Week 10 — Multiple regression I: one outcome, several causes

Business Forecasting

My code (.R) Notes

Back to Ecobici

Since W07, one regressor has forecast the daily trips:

$$\widehat{\text{trips}} = 16{,}893 + 723.6 \times \text{temp}, \qquad R^2 = 0.088$$

Temperature explains 8.8% of the day-to-day movement in trips. The fleet manager needs a better forecast.

What else moves the number of trips on a given day? Write down your guesses. For each one: can we measure it, and which way should it push trips?

From guesses to a regression

Take two guesses: days off and pollution.

- The outcome y: trips that day.
- The regressors: temperature, a day off (0 or 1), pollution that day (PM2.5, in µg/m^3).

The regression to write:

$$\text{trips}_i = \beta_0 + \beta_1 \, \text{temp}_i + \beta_2 \, \text{dayoff}_i + \beta_3 \, \text{pm25}_i + u_i$$

Each β is the change in trips when only its own variable changes and the others are held fixed. OLS finds all four numbers at once.

Build it by ticking boxes

Exercise 1 — A better forecast

`bikes` holds the 781 weekdays of W07–W09: `bikes$trips`, `bikes$temp`, `bikes$dayoff` (1 on a holiday or during the Christmas break) and `bikes$pm25` (pollution that day, PM2.5 in µg/m^3).

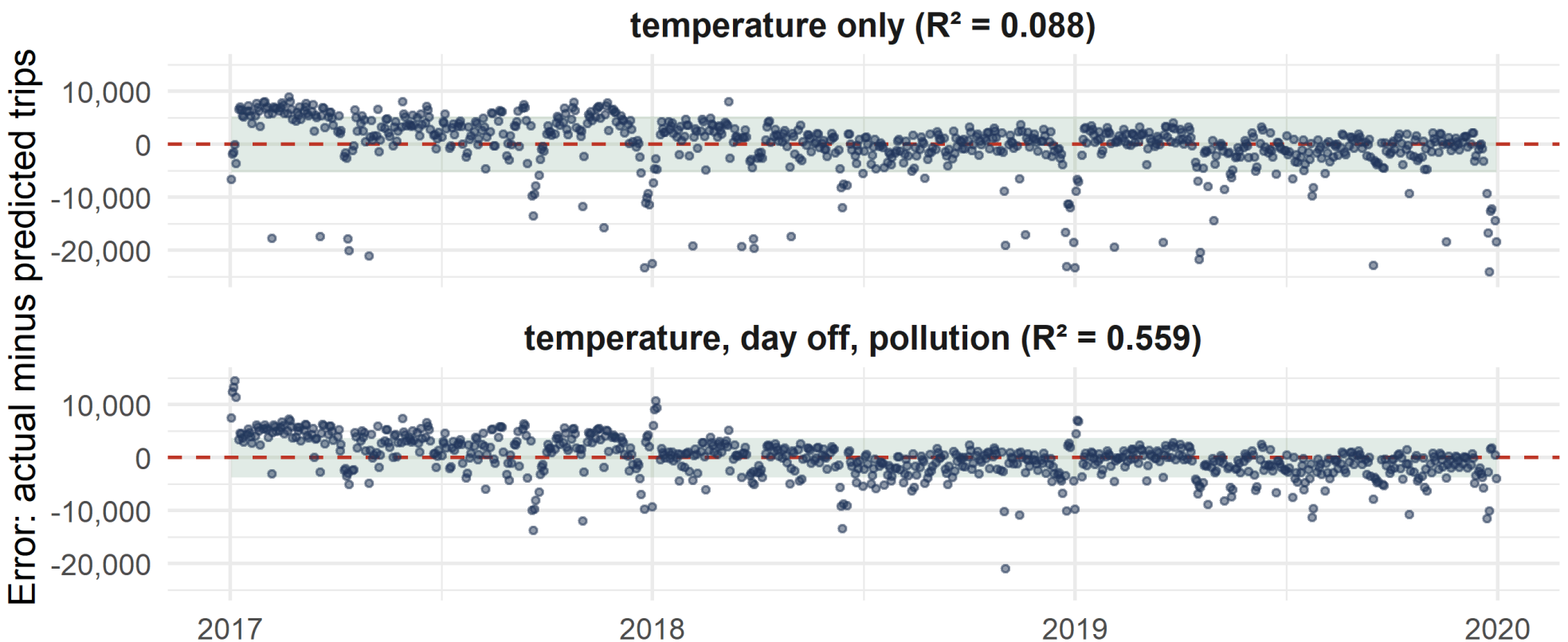
How much better does the forecast get with days off and pollution? How many trips should the fleet expect on a working day at 20 °C with PM2.5 at 25 µg/m^3?

1. Regress `bikes$trips` on `bikes$temp`, `bikes$dayoff` and `bikes$pm25`.
2. Compare R^2 and the residual standard error with temperature alone (0.088 and 5,302 trips).
3. Interpret the slope on `dayoff` and the slope on `pm25`. Does pollution help the forecast?
4. Forecast a working day at 20 °C with PM2.5 at 25 µg/m^3.

Exercise | Start Over | Show Hint | Run Code

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The errors, before and after



- Each dot is one day's error. The green band is plus or minus one typical miss.
- With temperature only, the holidays and the Christmas breaks sit far below zero. Once the days off are in the model, those big misses are gone and the band is narrower.

More regressors, compared
The same steps, six questions

How much is a year of education worth?

- The outcome: hourly wage, in dollars.
- What moves it? Education, experience, years with the employer, sex, marriage.

$$\text{wage}_i = \beta_0 + \beta_1\,\text{educ}_i + \beta_2\,\text{exper}_i + \beta_3\,\text{tenure}_i + \beta_4\,\text{female}_i + u_i$$

β_4 is the gap between a woman and a man with the same education, experience and tenure: the part of the pay gap these three cannot explain.

$$\widehat{\text{wage}}_i = \underset{(0.725)}{-1.57} + \underset{(0.0493)}{0.572}\,\text{educ}_i + \underset{(0.0116)}{0.0254}\,\text{exper}_i + \underset{(0.0212)}{0.141}\,\text{tenure}_i - \underset{(0.265)}{1.81}\,\text{female}_i$$

n = 526 workers, R^2 = 0.364. Standard errors in parentheses. US workers in 1976, `wooldridge` package (`wage1`).

With the same education, experience and tenure, a woman earns 1.81 dollars an hour less than a man ($T = -6.84$).

Does a convenience store charge more for a beer?

- The outcome: the shelf price of a pack of beer, in pesos.
- What moves it? The litres in the pack, the type of store, the brewer, the stores competing nearby, local income.

$$\text{price}_i = \beta_0 + \beta_1\,\text{litres}_i + \beta_2\,\text{convenience}_i + \beta_3\,\text{heineken}_i + \beta_4\,\text{rivals}_i + \beta_5\,\text{income}_i + u_i$$

β_2 asks whether a convenience store charges more than a supermarket for a pack of the same size and the same brewer, with the same competition nearby and the same local income.

$$\widehat{\text{price}}_i = \underset{(2.12)}{6.57} + \underset{(0.546)}{50.5}\,\text{litres}_i + \underset{(2.35)}{3.42}\,\text{convenience}_i + \underset{(1.29)}{9.83}\,\text{heineken}_i + \underset{(0.303)}{0.288}\,\text{rivals}_i - \underset{(0.199)}{0.0238}\,\text{income}_i$$

n = 2,252 packs, R^2 = 0.828. Standard errors in parentheses. PROFECO, Quién es Quién en los Precios, November 2025; income from INEGI, Encuesta Intercensal 2025.

A convenience store charges 3.42 pesos more for the same pack, but $T = 1.46$: not significant at 5%. The brewer matters more: a Heineken brand costs 9.83 pesos more than a Grupo Modelo brand in a pack of the same size.

What does a star player earn?

- The outcome: a basketball player's salary, in thousands of dollars a year.
- What moves it? Points, rebounds and assists per game, years in the league, being picked as an All-Star.

$$\text{wage}_i = \beta_0 + \beta_1\,\text{points}_i + \beta_2\,\text{rebounds}_i + \beta_3\,\text{assists}_i + \beta_4\,\text{exper}_i + \beta_5\,\text{allstar}_i + u_i$$

β_5 asks what the All-Star label adds for players with the same points, rebounds, assists and years in the league.

$$\widehat{\text{wage}}_i = \underset{(107)}{-163} + \underset{(11.9)}{76.7}\,\text{points}_i + \underset{(18.6)}{80.2}\,\text{rebounds}_i + \underset{(25.2)}{8.82}\,\text{assists}_i + \underset{(12.5)}{82.8}\,\text{exper}_i + \underset{(163)}{54.5}\,\text{allstar}_i$$

n = 269 players, R^2 = 0.551. Standard errors in parentheses. NBA salaries in the 1994-95 season, `wooldridge` package (`nbasal`).

One more point per game adds 76.7 thousand dollars a year. The All-Star label adds 54.5 thousand, but $T = 0.33$: not significant. Players with the same numbers earn about the same, All-Star or not.

What makes someone like their date?

- The outcome: after a four-minute speed date, how much a person liked their date, from 1 to 10.
- What moves it? How attractive, intelligent, fun and ambitious they found the date, and how much they share interests, each from 1 to 10.

$$\text{like}_i = \beta_0 + \beta_1\,\text{attractive}_i + \beta_2\,\text{intelligent}_i + \beta_3\,\text{fun}_i + \beta_4\,\text{ambitious}_i + \beta_5\,\text{shared}_i + u_i$$

β_1 asks how much more a person likes a date they find one point more attractive, comparing dates rated the same on intelligence, fun, ambition and shared interests. A dating app can use it to decide what to show first on a profile.

$$\widehat{\text{like}}_i = \underset{(0.297)}{0.304} + \underset{(0.0344)}{0.390}\,\text{attractive}_i + \underset{(0.0456)}{0.165}\,\text{intelligent}_i + \underset{(0.0369)}{0.232}\,\text{fun}_i - \underset{(0.0361)}{0.0231}\,\text{ambitious}_i + \underset{(0.0283)}{0.181}\,\text{shared}_i$$

n = 476 ratings, R^2 = 0.618. Standard errors in parentheses. Speed-dating experiment with Columbia University graduate students: 276 dates, each rated by both people.

`Lock5Data` package (`SpeedDating`), from Gelman and Hill (2007).

Looks have the largest slope: one more point adds 0.39 points of liking. Ambition adds nothing once the other four are held fixed ($T = -0.64$), although alone it moves with liking (correlation 0.35).

Does a CEO's pay follow the firm's results?

- The outcome: the CEO's salary, in thousands of dollars.
- What moves it? The firm's return on equity, its sales, its stock return.

$$\text{salary}_i = \beta_0 + \beta_1\,\text{roe}_i + \beta_2\,\text{sales}_i + \beta_3\,\text{ros}_i + u_i$$

β_1 asks whether a better-run firm pays its CEO more, comparing firms of the same size.

$$\widehat{\text{salary}}_i = \underset{(228)}{864} + \underset{(11.5)}{22.0}\,\text{roe}_i + \underset{(0.00895)}{0.0155}\,\text{sales}_i - \underset{(1.45)}{1.11}\,\text{ros}_i$$

n = 209 CEOs, R^2 = 0.032. Standard errors in parentheses. US CEOs in 1990, `wooldridge` package (`ceosal1`).

One more point of return on equity adds 22.0 thousand dollars of salary ($T = 1.91$, not significant at 5%). With $R^2 = 0.032$, the firm's results explain almost none of CEO pay.

Estimate all six

Exercise 2 — What is an extra bedroom worth?

`houses` holds 88 house sales from the `wooldridge` package (`hprice1`):

- `houses$price` — sale price, thousands of dollars
- `houses$bdrms` — number of bedrooms
- `houses$sqrft` — size of the house, square feet

1. Regress `houses$price` on `houses$bdrms`. How much is one more bedroom worth?
2. Regress `houses$price` on `houses$bdrms` and `houses$sqrft`. How much is one more bedroom worth now, for a house of the same size?
3. Compute the correlation between `houses$bdrms` and `houses$sqrft`. Why did the bedroom slope fall?

Exercise | Start Over | Show Hint | Run Code

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Holding size fixed

Model	Bedroom slope	Size slope	R^2
price on bedrooms	62.0		0.258
price on bedrooms and size	15.2	0.128	0.632

- An appraiser who prices bedrooms alone pays for size twice.
- With size held fixed, an extra bedroom is a smaller house cut into more rooms.

With the lot and the style held fixed too:

$$\widehat{\text{price}}_i = \underset{(29.6)}{-24.1} + \underset{(9.52)}{11.0}\,\text{bdrms}_i + \underset{(0.0133)}{0.124}\,\text{sqrft}_i + \underset{(0.000643)}{0.00208}\,\text{lotsize}_i + \underset{(14.6)}{13.7}\,\text{colonial}_i$$

n = 88 houses, R^2 = 0.676. Standard errors in parentheses. House sales in the Boston area, 1990, `wooldridge` package (`hprice1`).

One more bedroom adds 11.0 thousand dollars, but $T = 1.16$: not significant. Each extra square foot adds 124 dollars.

The rest of this week asks the same question of Mexican data from October 2025: how late will a flight land, once the other causes are held fixed?

How late will this flight land?

You plan the day for an airline at Mexico City airport (AICM). A flight from Monterrey left 40 minutes late. The gate, the ground crew and the passengers with a connection all need one number: how many minutes late will it land?

What moves the arrival delay? Write down your guesses. For each one: can we measure it, and which way should it push the delay?

Each cause needs its own number, measured with the others held fixed.

The data: arrivals at AICM

`flights` holds 11,887 flights that landed at Mexico City airport (AICM) in October 2025, from 111 airports. One row is one flight.

Column	What it is
`arr_delay`	minutes late at arrival (negative = early)
`dep_delay`	minutes late at departure (negative = early)
`sched_time`	scheduled flight time, minutes
`origin`, `code`, `country`, `intl`	where the flight comes from; `intl` is 1 for a flight from outside Mexico
`airline`	the airline
`date`, `weekday`, `hour`	day and hour of the scheduled arrival
`rain`, `wind`, `temp`	weather at AICM in that hour: rain (mm), wind (km/h), temperature (°C)

Schedules include slack, so most flights land before the scheduled time: only 14% land late.

Sources: arrival records of a public flight tracker; weather from the Open-Meteo historical archive. 359 records with extreme or impossible times are left out.

1 · How OLS finds the coefficients

Before the flights: how OLS finds several coefficients at once. The same rule as W07, the smallest sum of squared residuals.

Which three numbers?

A forecast for each flight needs three numbers:

$$\widehat{\text{arr_delay}}_i = b_0 + b_1 \, \text{dep_delay}_i + b_2 \, \text{sched_time}_i$$

Any three numbers give a forecast for each of the 11,887 flights, and an error for each: $e_i = \text{arr_delay}_i - \widehat{\text{arr_delay}}_i$.

Guess	b_0	b_1	b_2	Monterrey flight	SSE, millions	Typical miss
every flight lands on time	0	0	0	0	11.65	31.3
every flight lands at the average delay	−14.2	0	0	−14.2	9.25	27.9
30 minutes early, plus 0.8 of the departure delay	−30	0.8	0	2	2.39	14.2

$SSE = \sum e_i^2$, the sum of the squared errors. Typical miss $= \sqrt{SSE/n}$, in minutes. The Monterrey flight left 40 minutes late and is scheduled for 100 minutes.

The best three numbers bring the SSE down to 2.14 million, a typical miss of 13.4 minutes. Which three are they?

Still least squares

$$SSE(b_0, b_1, b_2) = \sum_{i=1}^{n} (\text{arr_delay}_i - b_0 - b_1 \, \text{dep_delay}_i - b_2 \, \text{sched_time}_i)^2$$

Write y_i for arr_delay_i, x_{i1} for dep_delay_i and x_{i2} for sched_time_i:

$$SSE(b_0, b_1, b_2) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2})^2$$

- OLS picks the three numbers that make SSE as small as possible.
- With one regressor this gave the W07 formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$.
- With two, it can still be solved by hand. With more, the answer is easiest to write with matrices.

Solve it by hand, without matrices

The flights as matrices

Stack the 11,887 flights, one row per flight. Hover over an entry to see what it is.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} -15 \\ -14 \\ \vdots \\ 99 \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix} = \begin{bmatrix} 1 & 6 & 99 \\ 1 & 10 & 221 \\ \vdots & \vdots & \vdots \\ 1 & 132 & 94 \end{bmatrix}$$

- $\mathbf{y}$ is $n \times 1$: the outcome, `arr_delay`, one entry per flight.
- $\mathbf{X}$ is $n \times 3$: a column of 1s for the intercept, then one column per regressor, `dep_delay` and `sched_time`.
- x_{ij} is regressor j of flight i: the row is the flight, the column is the regressor. $x_{21} = 10$: flight 2 left 10 minutes late.

The regression in matrix form

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}:\, n\times 1} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}}_{\mathbf{X}:\, n\times(k+1)} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix}}_{\boldsymbol{\beta}:\, (k+1)\times 1} + \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}}_{\mathbf{u}:\, n\times 1}$$

- One equation for all 11,887 flights at once.
- With k regressors, $\mathbf{X}$ has a column of 1s and then one column per regressor: $k+1$ columns. $\boldsymbol{\beta}$ has $k+1$ entries.
- The flights have $k = 2$: x_{i1} is `dep_delay` and x_{i2} is `sched_time`.

What $\mathbf{X}\boldsymbol{\beta}$ gives, row by row

The errors, as a function of $\mathbf{b}$

Pick any guess $\mathbf{b} = (b_0, b_1, \ldots, b_k)$. Every flight gets a forecast and an error.

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \begin{bmatrix} b_0 + b_1 x_{11} + \cdots + b_k x_{1k} \\ \vdots \\ b_0 + b_1 x_{n1} + \cdots + b_k x_{nk} \end{bmatrix}$$

$$\mathbf{e}(\mathbf{b}) = \mathbf{y} - \mathbf{X}\mathbf{b} = \begin{bmatrix} y_1 - b_0 - b_1 x_{11} - \cdots - b_k x_{1k} \\ \vdots \\ y_n - b_0 - b_1 x_{n1} - \cdots - b_k x_{nk} \end{bmatrix}$$

$$SSE(\mathbf{b}) = \sum_i e_i^2 = \begin{bmatrix} e_1 & \cdots & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \mathbf{e}^\top \mathbf{e}$$

$$\boxed{SSE(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b})}$$

The forecasts
Row i of $\mathbf{X}$ times $\mathbf{b}$: the forecast for flight i.

The errors
Each flight's outcome minus its forecast. Change $\mathbf{b}$ and every error changes.

The sum of squares
A row times a column: $e_1 e_1 + \cdots + e_n e_n$.

A function of $\mathbf{b}$ only
$\mathbf{y}$ and $\mathbf{X}$ are the data and stay fixed. OLS picks the $\mathbf{b}$ that makes this smallest.

Take the derivative

At the smallest SSE, the derivative with respect to $\mathbf{b}$ is zero, as in W07. Here it is in matrices.

$$SSE(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$$

$$= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{b} - \mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\,\mathbf{b}$$

$$= \mathbf{y}^\top \mathbf{y} - 2\,\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\,\mathbf{b}$$

$$\frac{\partial SSE}{\partial \mathbf{b}} = -2\,\mathbf{X}^\top \mathbf{y} + 2\,\mathbf{X}^\top \mathbf{X}\,\mathbf{b}$$

$$-2\,\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0} \;\Rightarrow\; \boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}}$$

Start from the errors
The sum of squared errors, as a function of $\mathbf{b}$.

Multiply out
Like $(a - c)(a - c)$, with $(\mathbf{X}\mathbf{b})^\top = \mathbf{b}^\top \mathbf{X}^\top$.

The two middle terms are equal
$\mathbf{y}^\top \mathbf{X}\mathbf{b}$ is one number, so it equals its own transpose, $\mathbf{b}^\top \mathbf{X}^\top \mathbf{y}$.

Derivative in $\mathbf{b}$
One derivative per coefficient, stacked in a column. As with one number: $-2bc$ gives $-2c$, and ab^2 gives $2ab$.

Set it to zero, factor out $-2\mathbf{X}^\top$
$\mathbf{y} - \mathbf{X}\mathbf{b}$ is $\mathbf{e}$. At the OLS coefficients, $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

The condition behind OLS

Write $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$ out. Row j of $\mathbf{X}^\top$ is column j of $\mathbf{X}$: one regressor, across all flights.

$$\mathbf{X}^\top \mathbf{e} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{n1} \\ \vdots & \vdots & & \vdots \\ x_{1k} & x_{2k} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} \sum_i e_i \\ \sum_i x_{i1} e_i \\ \vdots \\ \sum_i x_{ik} e_i \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- The first row: $\sum_i e_i = 0$. The errors average zero.
- Every other row: $\sum_i x_{ij} e_i = 0$. With errors that average zero, this says the errors are uncorrelated with regressor j.
- Row j is the derivative for one coefficient, $\partial SSE / \partial b_j = -2 \sum_i x_{ij} e_i$, set to zero.

OLS stops only when no regressor can explain anything left in the errors: if the errors still moved with the departure delay, a different b_1 would cut the SSE.

Check it in R

From the condition to the formula

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}$$

$$\mathbf{X}^\top \left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} \right) = \mathbf{0}$$

$$\mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X}\, \hat{\boldsymbol{\beta}} = \mathbf{0}$$

$$\mathbf{X}^\top \mathbf{X}\, \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$$

$$\hat{\boldsymbol{\beta}} = \boxed{\left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{y}}$$

Start from the condition
At the OLS coefficients, the errors move with no regressor.

Plug in the errors
At the minimum, $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$.

Multiply out
$\mathbf{X}^\top$ multiplies each part of the bracket.

Move one term across
The normal equations: one equation per coefficient.

Undo $\mathbf{X}^\top \mathbf{X}$
Multiply both sides by its inverse, the matrix version of dividing. This is the formula on your formula sheet.

With one regressor and no intercept, $\mathbf{X}^\top \mathbf{X} = \sum x_i^2$ and $\mathbf{X}^\top \mathbf{y} = \sum x_i y_i$, so $\hat{\beta} = \sum x_i y_i / \sum x_i^2$: how x moves with y, divided by how x moves with itself. With many regressors it is the same division, done with matrices. `lm()` does it for you.

Four flights, by hand | See the matrices move

Full rank: no column can be a copy

- $\mathbf{X}$ must have full rank: no column can be built exactly from the others.
- If one can, $(\mathbf{X}^\top \mathbf{X})^{-1}$ does not exist, and OLS has no single answer.

A colleague adds the scheduled time in hours, next to the scheduled time in minutes. What happens if both go into the same regression?

Near-copies: collinearity

Exercise 3 — Minutes and hours

1. Add `flights$sched_hours`, the scheduled time in hours: `flights$sched_time / 60`.
2. Regress `flights$arr_delay` on `flights$dep_delay`, `flights$sched_time` and `flights$sched_hours`. What does `summary()` print for `sched_hours`? Why?

Exercise | Start Over | Show Solution | Run Code

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2 · Departure and arrival

One regressor first: how much of a late departure is still there at landing?

Exercise 4 — Late out, late in

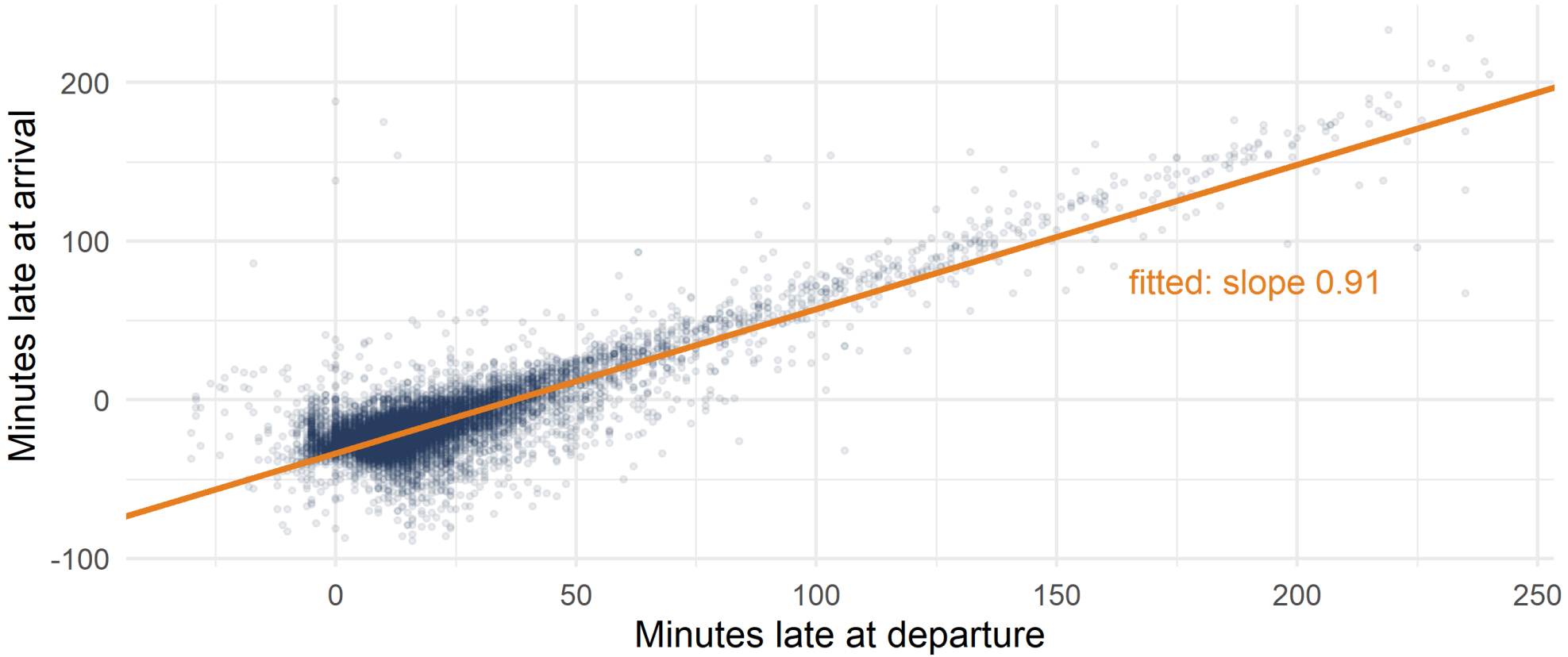
How many minutes late does a flight land for each minute it leaves late? How early does a flight land if it leaves on time?

1. Regress `flights$arr_delay` on `flights$dep_delay`.
2. Interpret the slope and the intercept, in minutes.
3. How much of the variation in arrival delays does the departure delay explain?

Exercise | Start Over | Show Solution | Run Code

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Most of a late departure stays



- $R^2 = 0.755$. The rest is the route, the airline, the weather and luck.

3 · A second regressor

Two causes in one model, and what “held fixed” means.

Do long flights make up more time?

A long flight has more time in the air, and more slack in its schedule, to make up a late departure.

$$\text{arr_delay}_i = \beta_0 + \beta_1 \, \text{dep_delay}_i + \beta_2 \, \text{sched_time}_i + u_i$$

- `sched_time` runs from 40 minutes to almost 16 hours. From Monterrey it is about 100 minutes; from Madrid about 720.
- β_2 compares two flights that left equally late, one with a longer schedule.

Exercise 5 — Does a long flight make up time?

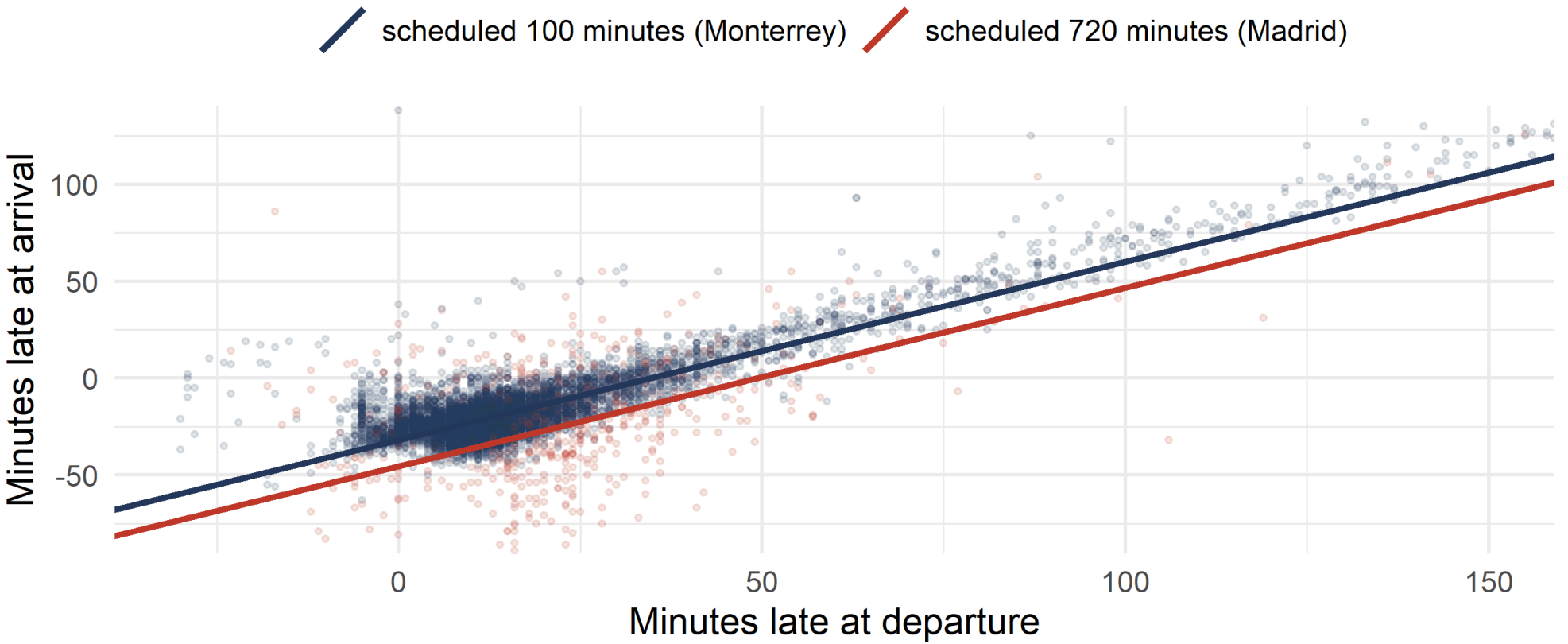
For two flights that left equally late, does the one with the longer schedule land earlier? By how many minutes per hour of schedule?

1. Regress `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Interpret both slopes, in minutes. Convert the `sched_time` slope to one extra hour of scheduled flight.
3. Did the `dep_delay` slope move from Exercise 4? How much did R^2 rise?

Exercise | Start Over | Show Hint | Run Code

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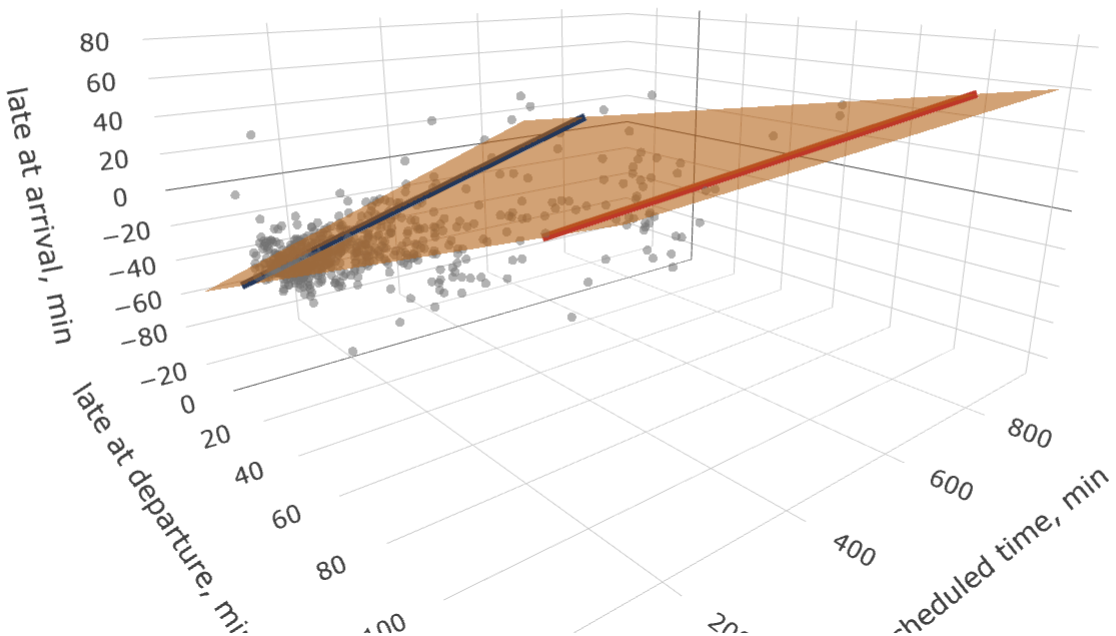
Two flights, two lines



- Two lines with the same slope, 0.92, drawn at exactly 100 scheduled minutes (Monterrey) and 720 (Madrid). The dots are the flights scheduled 3 hours or less, and 10 hours or more.
- The lines are 13.5 minutes apart: 620 minutes $\times$ 0.0217.
- Holding the scheduled time fixed means moving along one line.
- A 0/1 regressor, like `dayoff` for Ecobici, works the same way with only two values: it moves the whole line up or down.

Two regressors: a plane

With one regressor, the fitted values lie on a line. With the scheduled time added, they lie on a flat plane. Drag the picture to turn it.



- $\widehat{\text{arr_delay}} = -29.6 + 0.922\,\text{dep_delay} - 0.0217\,\text{sched_time}$: three numbers, one plane.
- Along the departure axis the plane rises 0.92 minutes per minute late. Along the scheduled-time axis it falls 0.022 minutes per scheduled minute.
- The blue and red lines are the 100- and 720-minute flights of “Two flights, two lines”: two slices of the plane.

One regressor: the line

Three or more regressors: no picture

Regressors	The fitted values lie on	Picture
1	a line	2D: the scatter and its line
2	a flat plane	3D: the plane you just turned
3 or more	a flat surface in 4 or more dimensions	none

Add the rain at AICM, in mm in the hour:

$$\widehat{\text{arr_delay}}_i = \underset{(0.215)}{-29.6} + \underset{(0.00464)}{0.922}\ \text{dep_delay}_i - \underset{(0.000804)}{0.0217}\ \text{sched_time}_i + \underset{(0.146)}{0.0524}\,\text{rain}_i$$

n = 11,887 flights, R^2 = 0.769. Standard errors in parentheses.

- No picture, but the formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ works the same with 3 columns or 30.
- Each slope still reads the same way: how $\hat{y}$ moves when one regressor moves and the others stay fixed.

Exercise 6 — Two forecasts

Two flights leave 40 minutes late: one from Monterrey, scheduled for 100 minutes, and one from Madrid, scheduled for 720 minutes. The gate team needs the expected arrival delay of each.

1. Compute both forecasts by hand from the coefficients of the regression of `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Compute them again with `predict()`.

Exercise | Start Over | Show Hint | Run Code

```
1
```

4 · When a new regressor moves a slope

Adding the scheduled time barely moved the departure slope. When does a new regressor move one?

Does pollution keep riders away?

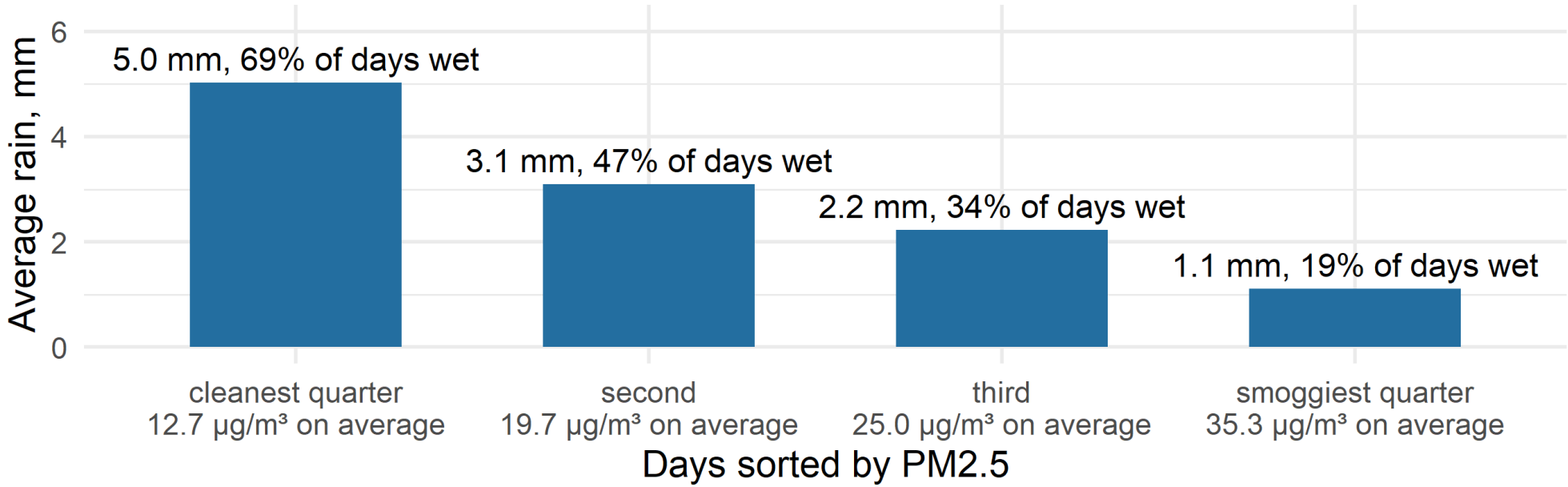
Back to Ecobici. On smoggy days, do fewer people ride? The city wants to know how many trips one more $\mu g/m^3$ of PM2.5 costs.

Before you fit anything, think about one more thing that changes from day to day: the rain.

Smoggy days: wet or dry?

`bikes$rain` is the rain that day in Mexico City, in mm. It is a continuous variable, not a 0/1 dummy: 0 on a dry day, up to 31 mm on the wettest.

Are smoggy days typically wet or dry?

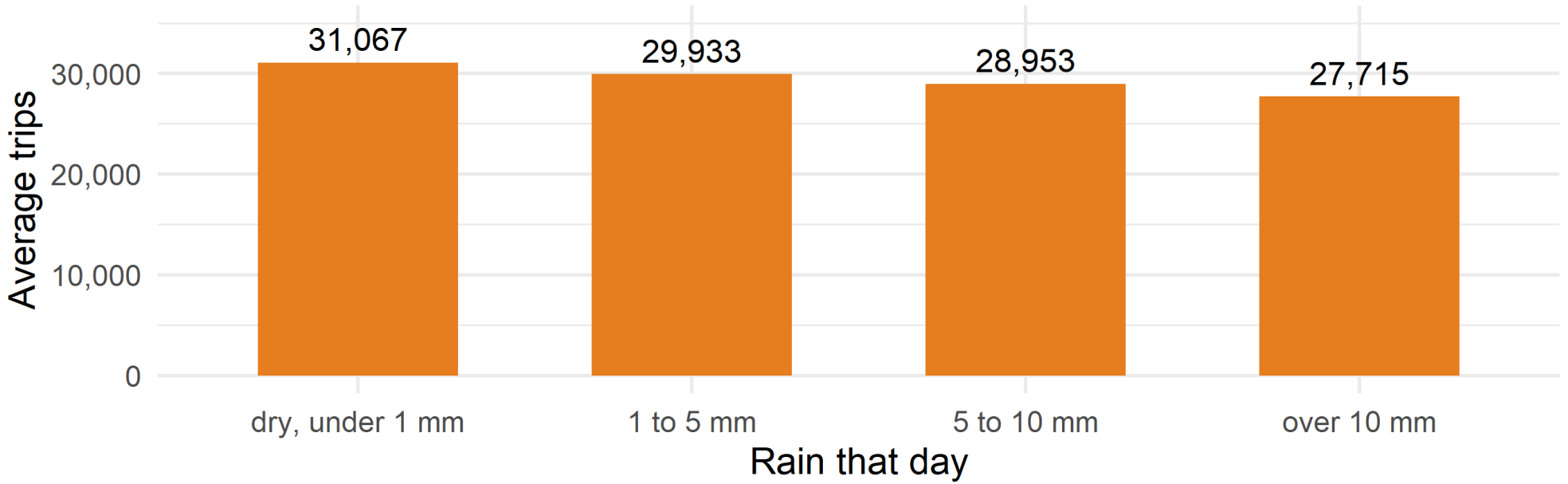


Smoggy days are mostly dry: rain washes the air.

So a regression of trips on PM2.5 alone compares smoggy days, mostly dry, with clean days, often wet.

Rain keeps riders at home

But rain also moves trips.



Working days only, so the holidays do not blur the picture.

So the PM2.5 slope alone partly measures the rain. The clean days lose riders to the rain, and that makes the gap between clean and smoggy days look smaller than the smog alone would make it.

Separate smog from rain

Compare days with the same amount of rain but different smog. Then rain cannot explain the difference in trips: what is left is the smog.

Predict: with rain held fixed, will the smog effect be bigger, smaller, or the same as with PM2.5 alone?

Exercise 7 — Pollution and rain

`bikes` as in Exercise 1, plus `bikes$rain`: the rain that day in Mexico City, in mm.

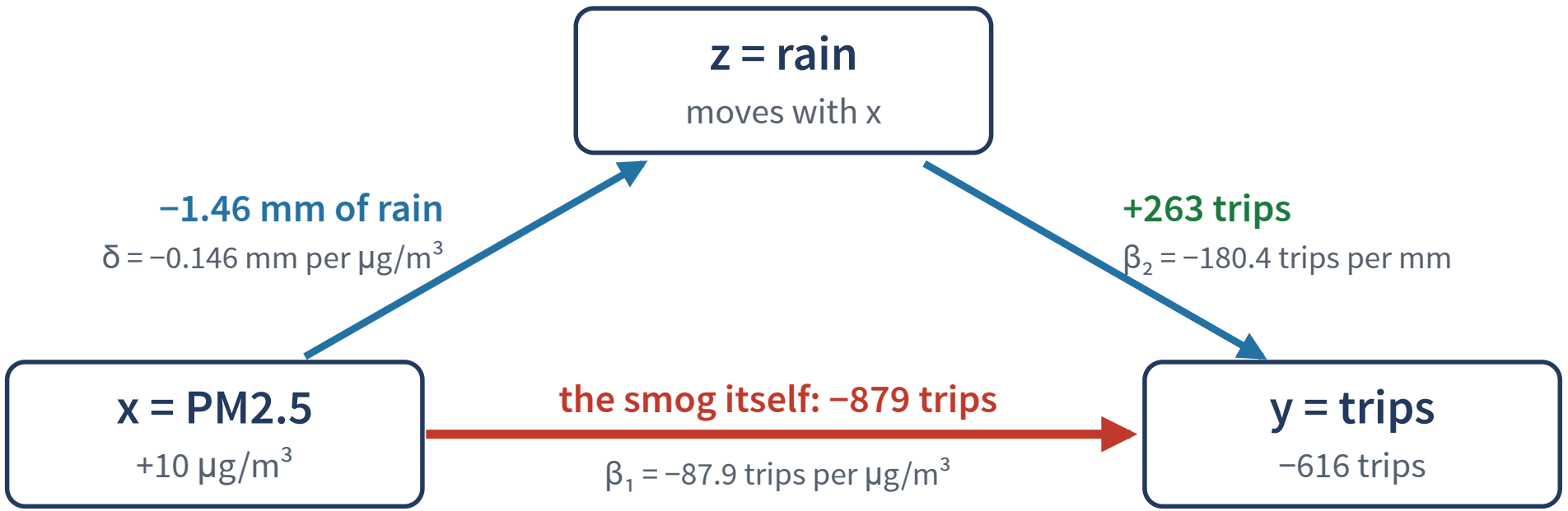
1. Regress `bikes$trips` on `bikes$pm25`. How many trips does one more µg/m^3 cost?
2. Add `bikes$rain` to the regression. How many trips does one more µg/m^3 cost now?
3. Compute the correlation of `bikes$pm25` with `bikes$rain`.
4. Did the smog effect get bigger or smaller with rain held fixed? Does it match your prediction?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Two roads from smog to trips

Less smog brings more riders. But less smog also comes with more rain, and rain keeps riders at home. Move the slider, then hold rain fixed.



A day **10** μg/m³ smoggier

PM2.5 alone | Rain held fixed

PM2.5 alone, both roads open: −879 + 263 = −616 trips. Per μg/m³: $\beta_1 + \beta_2\delta = -87.9 + 26.3 = -61.6$.

Alone, a slope compares high x with low x. They differ in x, and in everything that travels with x.

Holding z fixed compares cases with the same z, so z no longer travels with x.

The rule behind it

Held fixed, inside the formula

Two regressors: x = PM2.5 and z = rain, both measured from their means, so the intercept drops out. Hover over any sum.

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_x \\ \hat{\beta}_z \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} \sum x_i^2 & \sum x_i z_i \\ \sum x_i z_i & \sum z_i^2 \end{bmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{bmatrix} \sum x_i y_i \\ \sum z_i y_i \end{bmatrix}$$

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{D} \begin{bmatrix} \sum z_i^2 & -\sum x_i z_i \\ -\sum x_i z_i & \sum x_i^2 \end{bmatrix}$$

$$D = \sum x_i^2 \sum z_i^2 - \left(\sum x_i z_i \right)^2$$

$$\hat{\beta}_x = \frac{\sum z_i^2 \sum x_i y_i - \sum x_i z_i \sum z_i y_i}{D}$$

The OLS formula
The same formula as in section 1, with the columns x and z.

What goes in
Sums of squares and products. The red cross term: how x and z move together.

Invert it
The cross term comes back with a minus sign: the inverse discounts what x and z share.

What it gives for x
From the raw link of smog with trips, the formula takes out the part that runs through rain.

If x and z do not move together, $\sum x_i z_i = 0$ and $\hat{\beta}_x = \sum x_i y_i / \sum x_i^2$: the one-regressor slope of W07. Smog and rain do move together, so adding rain changed the smog slope.

Why it works: the part of x that z cannot predict

Exercise 8 — School, IQ and pay

`workers` holds 935 US men from the `wooldridge` package (`wage2`): `workers$wage`, monthly earnings in 1980 dollars; `workers$educ`, years of school; `workers$IQ`, IQ score.

How much more does one more year of school pay?

1. Regress `workers$wage` on `workers$educ`. How many dollars a month go with one more year of school?
2. Add `workers$IQ` to the regression. How many dollars now?
3. Compute the correlation of `workers$educ` with `workers$IQ`.
4. Comparing men with more and less school, what else are you comparing? Why did the slope move, and which way?

Exercise | Start Over | Show Solution | Run Code

```
1
```

5 · Fit with several regressors

More regressors always fit the sample at least as well. Is the gain worth it?

Can random numbers raise R^2?

So far R^2 has measured the fit. Each regressor we added raised it: in Exercise 1, days off and pollution took it from 0.088 to 0.559. That makes sense: the new regressors explain part of the errors the old model could not.

But R^2 has a weakness. Suppose we add a regressor of pure random numbers. Does R^2 still go up?

`bikes$noise` is such a column: random numbers with no link to trips. Add it to the regression of trips on temperature and days off. Predict: can R^2 rise?

Exercise 9 — Add pure noise

1. Regress `bikes$trips` on `bikes$temp` and `bikes$dayoff`. Note the R^2.
2. Add `bikes$noise` to the regression. Note the R^2 again. Did it rise?
3. Is `noise` significant at 5%?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Why R^2 can only rise

$$SSE_{\text{old}} = \min_{b_0,\, b_1,\, b_2} \sum_i (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2})^2$$

$$SSE_{\text{new}} = \min_{b_0,\ldots,b_3} \sum_i (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2} - b_3 x_{i3})^2$$

$$b_3 = 0 \;\Rightarrow\; \text{the old model, with } SSE_{\text{old}}$$

$$\boxed{SSE_{\text{new}} \le SSE_{\text{old}} \;\Rightarrow\; R^2_{\text{new}} \ge R^2_{\text{old}}}$$

The old model
OLS picks the smallest SSE with temperature and days off.

Add the noise column
One more coefficient to choose, b_3.

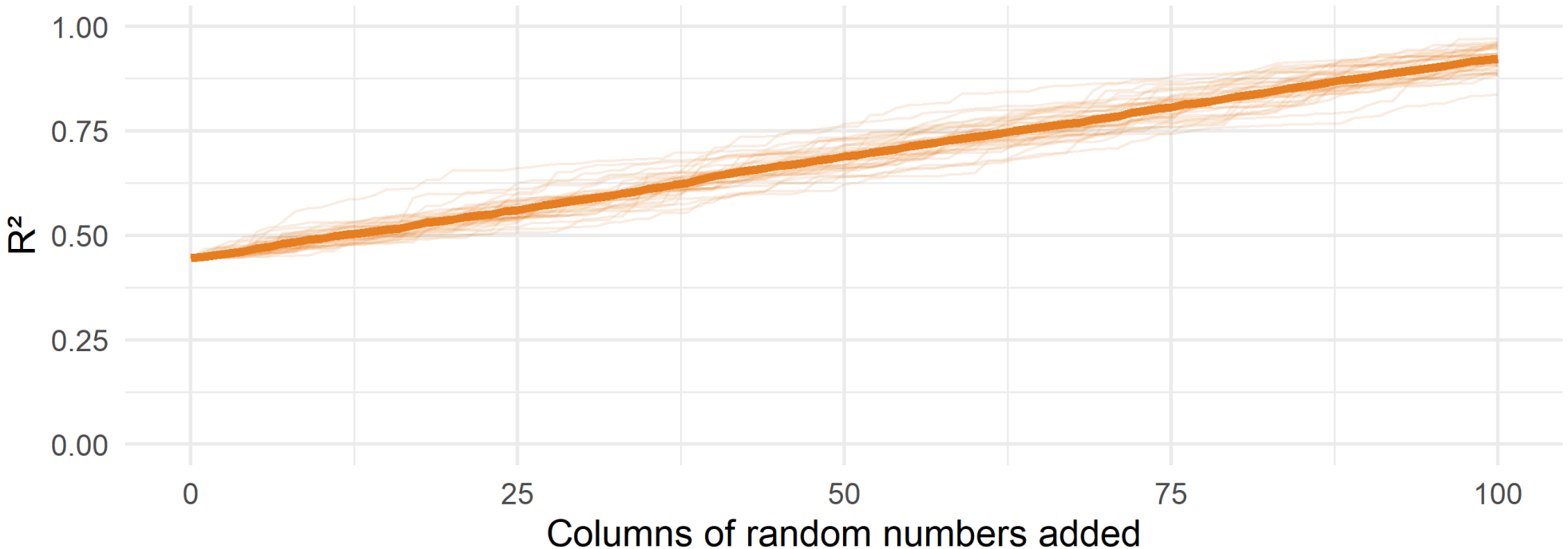
One choice is to ignore it
With its slope set to zero, the new column changes nothing.

The best can only be as good or better
OLS can always fall back on $b_3 = 0$. SST does not change, so $R^2 = 1 - SSE/SST$ cannot fall.

And it almost always rises. In a sample, random numbers are never exactly uncorrelated with the errors: in Exercise 9 the correlation of `bikes$noise` with the errors is -0.018. Tiny, but OLS uses it to cut the SSE a little.

100 columns of random numbers

Take 120 Ecobici days, trips on temperature and days off. Add random columns, one at a time, up to 100.



- Thin lines: 40 different sets of random columns. Thick line: their average.
- R^2 climbs from 0.45 to 0.92 with 100 columns of pure noise. With 117 columns it would reach 1: a perfect fit to noise.

Judged by R^2, the model with 100 random columns is the best one. That makes no sense.

Each coefficient needs a fee: keep a regressor only if what it adds to the fit pays for the coefficient it uses.

Adjusted R^2: a fee for each regressor

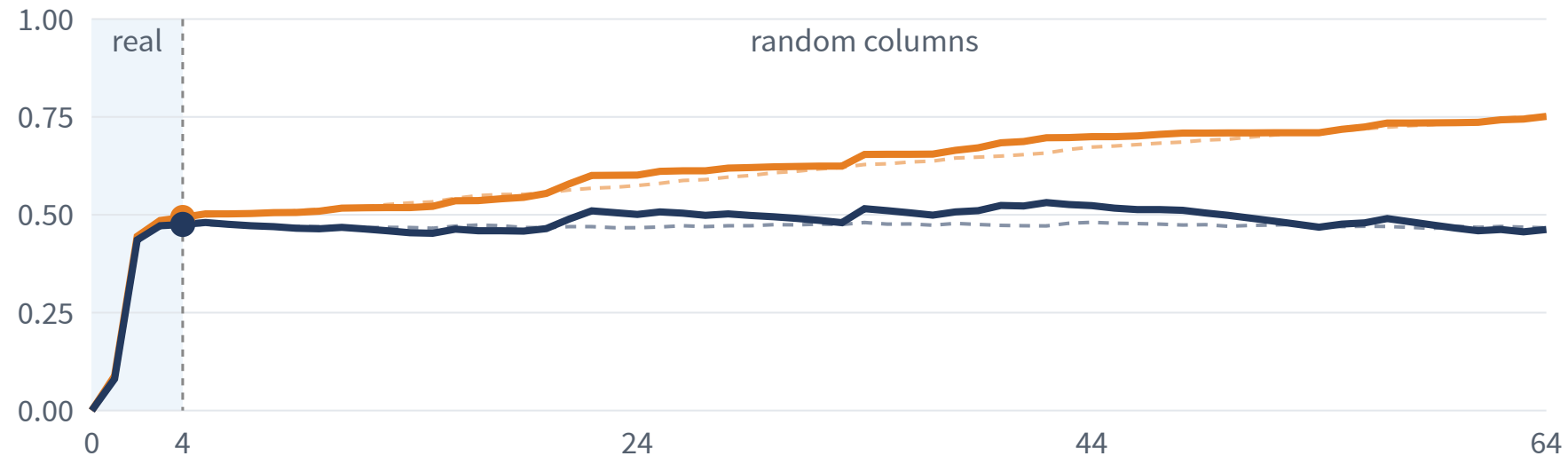
$$R^2 = 1 - \frac{SSE}{SST} \qquad \bar{R}^2 = 1 - \frac{SSE/(n-k-1)}{SST/(n-1)} = 1 - \frac{s^2}{\text{var}(y)}$$

- R^2 looks only at SSE, and every new coefficient lowers it.
- Adjusted R^2 divides SSE by $n-k-1$. Each new coefficient lowers $n-k-1$ by one: that is the fee.
- A new regressor raises adjusted R^2 only if it cuts SSE by more than the fee. This happens exactly when its $|T|$ is above 1.
- Adjusted R^2 is no longer exactly the share of the variation explained: it is that share, adjusted for the number of regressors.
- To compare models with different numbers of regressors, use adjusted R^2.

All 781 Ecobici days, trips on temperature and days off: s = 3,690 trips and the standard deviation of trips is 5,548, so adjusted R^2 is 1 − 3,690^2 / 5,548^2 = 0.558.

Watch both numbers

The same 120 Ecobici days. The first four regressors are real: temperature, day off, rain, PM2.5. After them come columns of random numbers. Move the slider and watch both numbers.



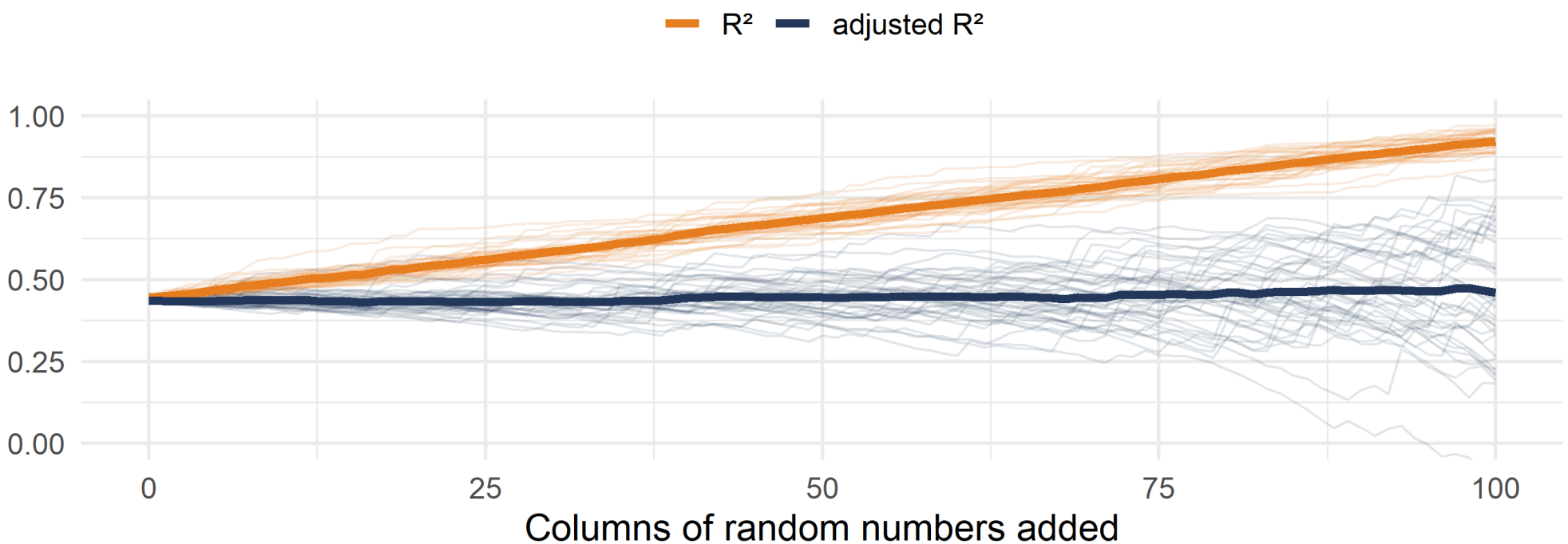
− + k = 4: temperature, day off, rain, PM2.5 New random columns (1 of 10)

$R^2 = 1 - SSE / SST = 1 - 1.569 / 3.095 = 0.493$

adjusted $R^2 = 1 - (SSE / (n - k - 1)) / (SST / (n - 1)) = 1 - (1.569 / 115) / (3.095 / 119) = 0.475$

SSE and SST in billions of squared trips. Solid lines: one set of random columns. Dashed lines: the average over 10 sets.

The fee at work



- The same random columns. On average, adjusted R^2 stays near 0.44: it is 0.46 with 100 columns. The fee cancels what the noise seems to add.
- Single draws swing more as the columns pile up: 120 days pay for up to 103 coefficients.

Exercise 10 — Noise, with the fee

1. Run the two regressions of Exercise 9 again: `bikes$trips` on `bikes$temp` and `bikes$dayoff`, then with `bikes$noise` added.
2. Compare the adjusted R^2 of the two. Which model wins now?
3. Check the rule: what is the $|T|$ of `noise`?

Exercise | Start Over | Show Hint | Run Code

```
1
```

R^2 or adjusted R^2?

- R^2 is the share of the variation in y that the model explains. Adding a regressor never lowers it, even a column of random numbers.
- Adjusted R^2 is that share adjusted for the number of regressors. It is no longer exactly the share explained: each coefficient pays a fee.
- A new regressor raises adjusted R^2 only when its $|T|$ is above 1.

To compare models that differ in the number of regressors, use adjusted R^2.

Does the weather matter?

One route only: the 682 flights from Guadalajara, all scheduled for about 90 minutes.

Once the departure delay is known, does the weather at AICM help forecast the arrival delay?

Exercise 11 — The weather and adjusted R^2

1. Keep the Guadalajara flights: `gdl <- flights[flights$code == "GDL", ]`. Regress `gdl$arr_delay` on `gdl$dep_delay`. Note the R^2 and the adjusted R^2.
2. Add `gdl$rain`. Note both again. Is `rain` significant at 5%?
3. Replace `rain` with `gdl$wind`. Note both again.
4. Which weather variable raised the adjusted R^2?

Exercise | Start Over | Show Solution | Run Code

```
1
```

What the weather adds

Rain lowers the adjusted R^2 and wind raises it a little. Neither is significant at 5%.

For the operations team: the departure delay carries the forecast, and a long schedule absorbs part of it. The weather at the airport adds almost nothing once the departure delay is known.

Do some airlines make up more time than others? That needs a regressor with several categories: week 11.

Exercise 12 — Brief the operations manager

Give the manager the expected arrival delay of flights that leave 0, 30 and 60 minutes late, from Monterrey (100 scheduled minutes) and from Madrid (720), with one picture and one sentence.

1. Use the regression of `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`.
2. Forecast the six delays with `predict()`.
3. Plot `flights$arr_delay` against `flights$dep_delay`.
4. Write one sentence for the manager.

Exercise | Start Over | Show Hint | Run Code

```
1
```

What you should be able to do

- Explain why one cause at a time misleads, and what “held fixed” means.
- Derive the OLS formula from the condition that the errors move with no regressor.
- Explain full rank, and what R does with an exact copy of a regressor.
- Fit a regression with several regressors and read each slope with the others held fixed.
- Read the coefficient of a 0/1 regressor as the gap between two parallel lines.
- Forecast with several regressors, by hand and with `predict()`.
- Say when adding a regressor moves another slope, and why.
- Read the two-regressor formula: the cross term S_{12} is how the other regressor is held fixed.
- Explain why R^2 always rises, and compare models with adjusted R^2.

Next week: regressors with several categories, such as the airline, and interactions, with PROFECO shelf prices: whether small local stores price to their neighbourhood, and what the January 2026 tax did to drink prices.

Go to the exam practice →

Practice for this week

Weekly exercise set: `W10_exercises.pdf`, with solutions in `W10_solutions.pdf`.

You can already solve exercises 2, 3 and 25, and most of 4, 5 and 6.

Not in these slides: 1 and 7(c) use regressors with several categories (week 11); 4(f), 5(i) and 6(b) use intervals for a coefficient or a forecast (week 12); 7(a)–(b) use a squared regressor, which this course does not cover.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
2	Grades on two GRE scores and a 0/1 regressor (`metricsgrades`)	R with the data
3	Cigarette tax and price move together (`cigdata`)	R with the data
25	Food spending on income and household size (ENIGH)	R with the data
5 (a)–(h)	Vote shares on four district traits (`congress`)	R with the data

Interactive versions: exam practice

Exam practice

Problems from this week's set. ← Back to the summary · ← Back to Practice for this week

Exam practice 1 — Grades and the GRE

[P3 #2]

`metricsgrades` holds 68 students of a graduate econometrics course, from the `probstats4econ` package.

- `metricsgrades$total` — final course grade, out of 100
- `metricsgrades$gre_quant`, `metricsgrades$gre_verbal` — GRE scores, out of 170
- `metricsgrades$domestic` — 1 if a US student, 0 otherwise

a. Compute the correlation matrix of the four variables. Which one has the largest correlation, in magnitude, with `total`?

b. Regress `total` on the other three variables.

c. Interpret the slope on `gre_quant`.

d. Predict the grade of a non-domestic student with 160 in quant and 150 in verbal.

e. What is the estimated standard deviation of the errors?

f. Test $H_0 : \beta_{\text{domestic}} = 0$ against $H_A : \beta_{\text{domestic}} \neq 0$ at the 10% level, using the output of `summary()`.

g. Drop `domestic` and fit again. What changes?

h. Put `domestic` back and drop `gre_verbal`. What happens to the significance of `domestic`, and why?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 2 — Cigarette taxes and prices

[P3 #3]

`cigdata` holds the 50 US states and Washington D.C. in 2019, from the `probstats4econ` package.

- `cigdata$cigsales` — packs sold per person per year
- `cigdata$cigtax` — state tax per pack, dollars
- `cigdata$cigprice` — total price per pack, dollars
- `cigdata$producer` — 1 if the state grows tobacco

a. Regress `cigsales` on `cigtax` and `producer`. Interpret the slopes and R^2.

b. Add `cigprice`. How does R^2 change?

c. What happens to the significance of the tax slope?

d. Use the correlation between `cigtax` and `cigprice` to explain (c).

e. Drop `cigtax` and keep `cigprice` and `producer`. Compare with (a) and (b).

f. Which model do you prefer, (`cigtax`, `producer`) or (`cigprice`, `producer`)? Why?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 3 — Mutual fund returns

[P3 #4]

`mutualfunds` holds 208 large-blend equity funds, from the `probstats4econ` package.

- `mutualfunds$return_10yr` — ten-year annualised return (0.08 = 8%)
- `mutualfunds$expense_ratio` — annual fee (0.005 = 0.5%)
- `mutualfunds$manager_tenure`, `mutualfunds$fund_age` — years
- `mutualfunds$load` — "Y" if the fund charges a sales fee, "N" otherwise

a. Add a column `hasload`, 1 if `load` is "Y" and 0 otherwise.

b. Regress `return_10yr` on `expense_ratio`, `manager_tenure`, `fund_age` and `hasload`.

c. Interpret R^2.

d. What does a 0.001 rise in `expense_ratio` do to the ten-year return?

e. Which slopes are significant at 5%?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 4 — District elections

[P3 #5]

`congress` holds the 476 US House district elections of 1990, from the `probstats4econ` package.

- `congress$demvoteshare` — share of the vote for the Democrat (0 to 1)
- `congress$democrat` — 1 if the Democrat won; `congress$lagdemocrat` — 1 if they won the election before
- `congress$medianincome` — median district income, dollars
- `congress$pcturban`, `congress$pctblack`, `congress$pcthighschl` — shares urban, Black, with a high-school degree

a. Draw a histogram of `demvoteshare`. One peak or two?

b. What fraction of the elections did Democrats win?

c. Compute the mean of each of the four district traits where Democrats won and where Republicans won.

d. For each trait, regress it on `democrat` and say, from the p-value, whether its mean differs between the two groups at the 5% level.

e. Regress `demvoteshare` on the four traits.

f. Interpret the income slope for a 1,000-dollar rise.

g. Interpret the urban slope for a rise of 10 percentage points.

h. Add `lagdemocrat`. What happens to R^2 and to the other slopes?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 5 — Out-of-pocket medical costs

[P3 #6]

`hrs` holds 6,052 unmarried people aged 50 and over, from the Health and Retirement Study (`probstats4econ` package). Use only the 3,983 with positive out-of-pocket costs.

- `hrs$outofpocket_costs` — medical costs paid out of pocket in 2000, dollars
- `hrs$age` — years
- `hrs$ins_none` — 1 if no health insurance; `hrs$ins_medicare` — 1 if Medicare
- `hrs$male` — 1 if male

a. Regress `outofpocket_costs` on `age`, `ins_none`, `ins_medicare` and `male`. Interpret the slopes on `age` and `ins_none`.

Exercise | Start Over | Show Solution | Run Code

```
1
```

📝 Exam practice 6 — A second regressor for food spending

[25]

`Hp` holds the 74,639 ENIGH households with positive income (Canvas, 02 Data, `Dataset_spending_income.Rda`).

- `Hp$food_spending` — monthly food spending, pesos
- `Hp$monthly_income` — monthly income, pesos
- `Hp$HHmembers` — people in the household

a. Regress `food_spending` on `monthly_income`. Report the slope and R^2.

b. Add `HHmembers`. Report the three coefficients, the two standard errors and R^2.

c. Interpret the slope on `HHmembers`.

d. How does the meaning of the income slope differ between (a) and (b)?

e. The income slope barely moved. Use the correlation between income and household size to explain why.

f. A supermarket chain sizes its stores from these two variables. Which matters more for food spending, and what is the evidence?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — optional

The Ecobici regression builder and the models it compares, the six questions in one table, the builder for all six, the rule for when a new regressor moves a slope, a slope as the part of x_1 that x_2 cannot predict, the algebra of omitted-variable bias, two more examples of it, collinearity, the fit with one regressor drawn as a line, least squares solved by hand, the matrix product row by row, the errors checked in R, and the matrix formula by hand.

Appendix — build the Ecobici regression

Tick the regressors the class proposed. The regression to estimate appears first; then show the estimates.

What moves the number of Ecobici trips? (781 weekdays, 2017-2019)

REGRESSORS

- ☐ **temp** temperature, °C
- ☐ **dayoff** 1 on a holiday or the Christmas break
- ☐ **monday** 1 on Mondays
- ☐ **friday** 1 on Fridays
- ☐ **pm25** pollution, PM2.5 in µg/m^3
- ☐ **rain** rain that day, mm

← Back to the guesses

THE REGRESSION TO ESTIMATE

$$\texttt{trips}_i = \beta_0 + u_i$$

Show the estimates | Clear

Appendix — how much better?

Regressors	R^2	adjusted R^2	Typical miss (trips)
temp	0.088	0.087	5,302
temp + dayoff	0.559	0.558	3,690
temp + dayoff + pm25	0.559	0.557	3,691
temp + dayoff + monday + friday	0.579	0.577	3,609
temp + dayoff + monday + friday + pm25	0.579	0.577	3,611

- The days off do most of the work: R^2 jumps from 0.088 to 0.559.
- Pollution adds nothing: adjusted R^2 falls. Mondays and Fridays add a little more.
- A better regression is a better forecast: the typical miss shrinks from 5,302 to about 3,600 trips.

← Back to the errors

Appendix — regression at work

The same steps work for any business question: name the outcome, list what moves it, write the regression, estimate it, read each slope with the others held fixed.

Business question	Data
What is an extra bedroom worth?	88 house sales
How much is a year of education worth? Is there a gender gap?	526 workers
Does a convenience store charge more for a beer?	2,252 beer packs
What does a star basketball player earn?	269 players
What makes someone like their date?	476 speed-date ratings
Does a CEO's pay follow the firm's results?	209 CEOs

Four come from the `wooldridge` package: `hprice1`, `wage1`, `nbasal`, `ceosal1`; the beer prices from PROFECO, Quién es Quién en los Precios; the speed dates from the `Lock5Data` package (`SpeedDating`).

← Back to the errors

Appendix — estimate them

Pick a question, tick the regressors, and compare the slopes and R^2 across models.

Houses: sale price

What is an extra bedroom worth? (88 house sales, Boston area, 1990 (wooldridge: hprice1))

REGRESSORS

- ☐ **bdrms** bedrooms
- ☐ **sqrft** size, square feet
- ☐ **lotsize** lot, square feet
- ☐ **colonial** 1 if colonial style

THE REGRESSION TO ESTIMATE

$$price_i = \beta_0 + u_i$$

Show the estimates | Clear

← Back to the CEOs

Appendix — the rule

New regressor	Correlation with PM2.5	Moves trips?	PM2.5 slope
temperature	−0.01	yes: +722 trips per °C	−61.6 → −60.5
rain	−0.29	yes: −180 trips per mm	−61.6 → −87.9

A slope moves only when the new regressor does both: it moves the outcome, and it moves with the old regressor. Leaving such a variable out is omitted-variable bias.

Temperature moves trips but not with pollution, so the PM2.5 slope stays. Rain does both:

$$\underbrace{-61.6}_{\text{alone}} = \underbrace{-87.9}_{\text{rain held fixed}} + \underbrace{(-180.4)}_{\text{trips per mm of rain}} \times \underbrace{(-0.146)}_{\text{mm of rain per } \mu\text{g/m}^3}$$

The algebra behind it | Another example: cold days or days off | Another example: the slack in long flights | ← Back to the two roads

Appendix — the algebra of omitted-variable bias

True model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. We fit y on x_1 alone.

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 (\delta_0 + \delta_1 x_1 + v) + u$$

$$y = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1)\, x_1 + (\beta_2 v + u)$$

$$\text{slope alone} = \boxed{\beta_1 + \beta_2\, \delta_1}$$

Start
Write the left-out regressor as a line in x_1. δ_1 says how they move together.

Plug it in
Replace x_2 in the true model.

Collect the terms in x_1
The slope on x_1 alone is the red part.

Both conditions
The bias $\beta_2 \delta_1$ is zero if x_2 does not move y ($\beta_2 = 0$) or does not move with x_1 ($\delta_1 = 0$).

In the Ecobici data, with x_1 = `pm25` and x_2 = `rain`, the identity holds exactly for the OLS estimates: $-87.94 + (-180.37) \times (-0.1460) = -61.62$. With x_1 = `temp` and x_2 = `dayoff`: $394.89 + (-15837.65) \times (-0.0208) = 723.55$. In the flights example, with x_1 = `sched_time` and x_2 = `dep_delay`: $-0.0217 + 0.9215 \times 0.0161 = -0.0069$.

← Back to the rule

Appendix — cold days or days off?

Back to Ecobici. With temperature alone, each extra degree adds 724 trips. In Exercise 1, with the days off and pollution added, it adds only 394.

The fleet manager plans bikes and staff for a cold working week. Which slope belongs in that plan?

Predict before you fit: which new regressor moved the temperature slope, the days off or pollution?

Appendix — cold days or days off, in R

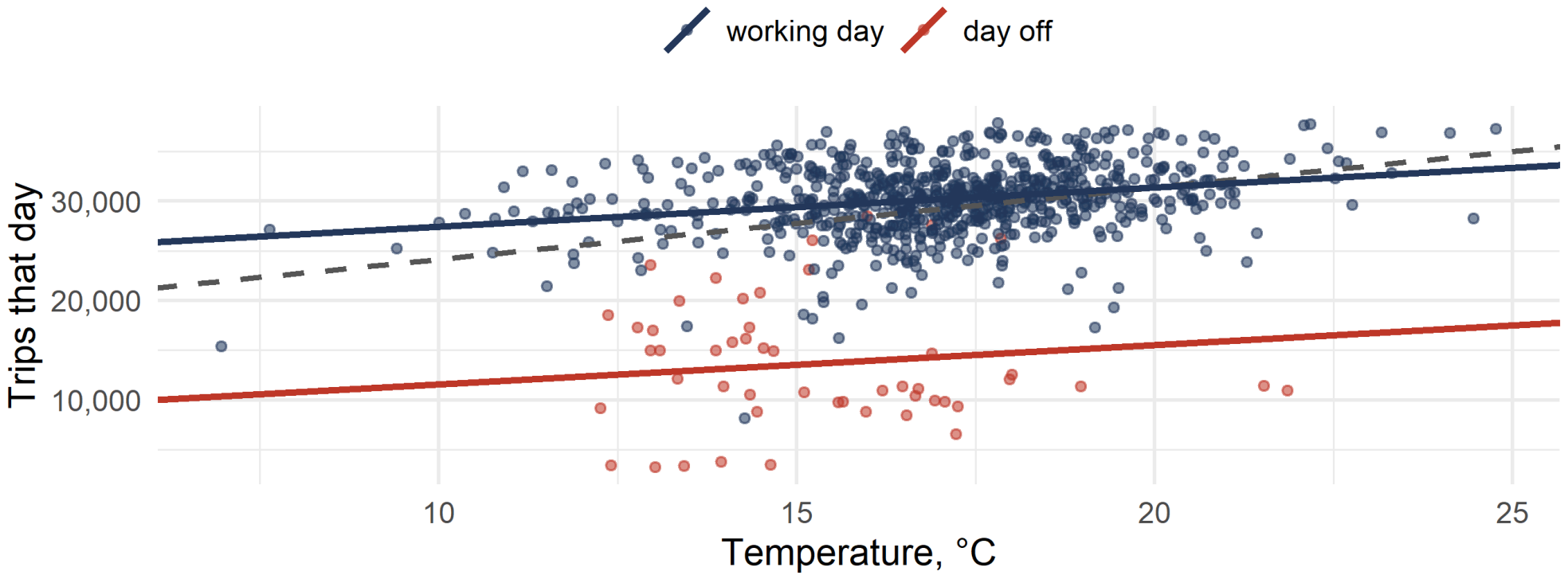
`bikes` as in Exercise 1: `bikes$trips`, `bikes$temp`, `bikes$dayoff` and `bikes$pm25`.

1. Regress `bikes$trips` on `bikes$temp` and `bikes$pm25`. Did the temperature slope move from 723.6?
2. Regress `bikes$trips` on `bikes$temp` and `bikes$dayoff`. Did it move now?
3. Compute the correlation of `bikes$temp` with `bikes$pm25`, and of `bikes$temp` with `bikes$dayoff`.
4. Why did one regressor move the temperature slope and the other not? Which slope belongs in the plan for a cold working week?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — Christmas is cold



- The red dots are the days off. They sit low, and most of them are the Christmas break, in the coldest weeks of the year: 15.3 °C on average, against 17.1 °C on working days.
- The dashed line, temperature alone, is steep: its cold end is pulled down by the empty holidays.
- The two solid lines hold the type of day fixed. Same slope, 395 trips per degree, 15,838 trips apart.
- A small correlation is enough when the left-out cause is strong: a day off removes about 15,838 trips.

← Back to the rule

Appendix — the slope uses only what is new in x_1

Write $x_1 = x$ and $x_2 = z$, measured from their means, with $S_{11} = \sum x_{i1}^2$, $S_{12} = \sum x_{i1} x_{i2}$, $S_{22} = \sum x_{i2}^2$, $S_{1y} = \sum x_{i1} y_i$ and $S_{2y} = \sum x_{i2} y_i$. The formula slide gives $\hat{\beta}_1 = (S_{22} S_{1y} - S_{12} S_{2y}) / (S_{11} S_{22} - S_{12}^2)$.

$$\hat{\beta}_1 = \frac{S_{1y} - \delta\, S_{2y}}{S_{11} - \delta\, S_{12}}, \qquad \delta = \frac{S_{12}}{S_{22}}$$

$$\hat{\beta}_1 = \boxed{\frac{\sum r_i \,(y_i - \bar{y})}{\sum r_i^2}}$$

$$r_i = (x_{i1} - \bar{x}_1) - \delta\,(x_{i2} - \bar{x}_2)$$

Divide top and bottom by S_{22}

δ is the slope of a regression of x_1 on x_2: how much x_1 moves with x_2.

Write the sums back out

The top is $\sum r_i (y_i - \bar{y})$. The bottom is $\sum r_i^2$, because $\delta^2 S_{22} = \delta S_{12}$. This is the W07 slope formula, with x_1 replaced by r.

r is the part of x_1 that x_2 cannot predict. Holding x_2 fixed means: keep only that part of x_1, and run the one-regressor regression of W07 on it.

← Back to the formula

Appendix — held fixed, in one-regressor regressions

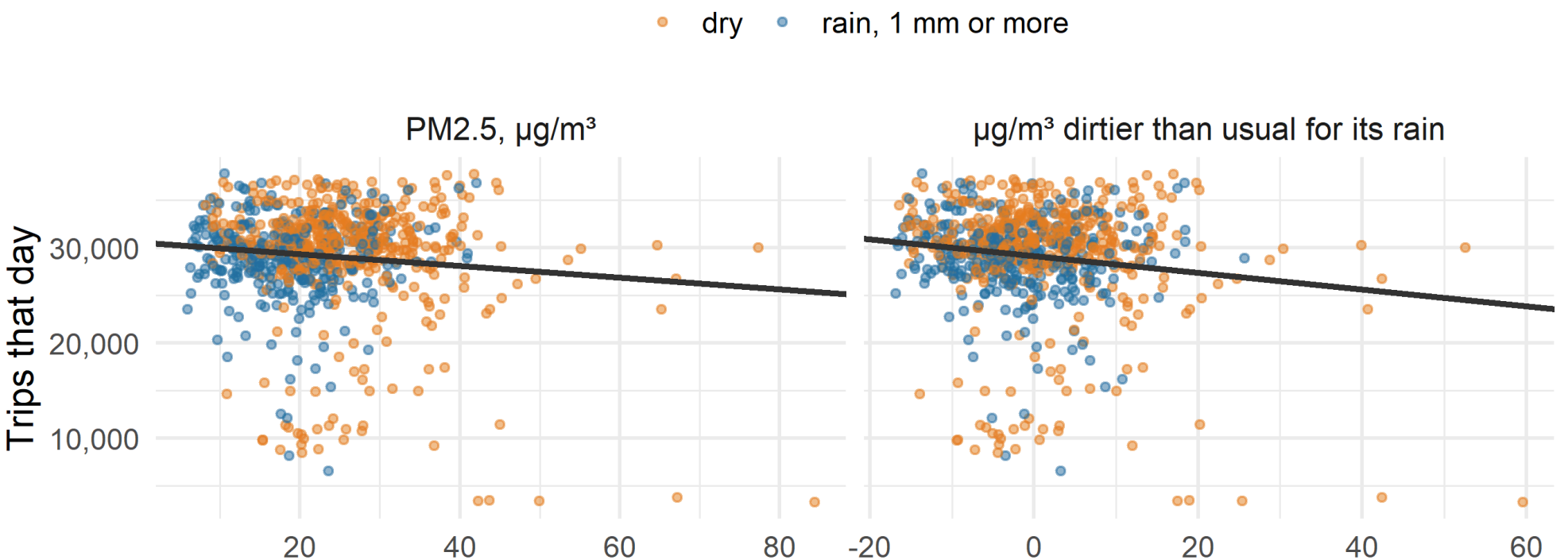
With rain held fixed, one more μg/m^3 of PM2.5 costs 87.9 trips (Exercise 7). Get the same number from regressions with one regressor each.

1. Regress `bikes$pm25` on `bikes$rain`. Save its residuals as `bikes$pm25_r`: how much dirtier or cleaner each day was than usual for its rain.
2. Regress `bikes$trips` on `bikes$pm25_r`. Compare its slope with the `pm25` slope of the regression of `bikes$trips` on `bikes$pm25` and `bikes$rain`.
3. Compute the correlation of `bikes$pm25_r` with `bikes$rain`. What does it say?
4. Same steps for `houses`: regress `houses$bdrms` on `houses$sqrft`, save the residuals as `houses$bdrms_r`, and regress `houses$price` on `houses$bdrms_r`. Compare with the bedroom slope of Exercise 2, with size held fixed.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Appendix — what the rain took out

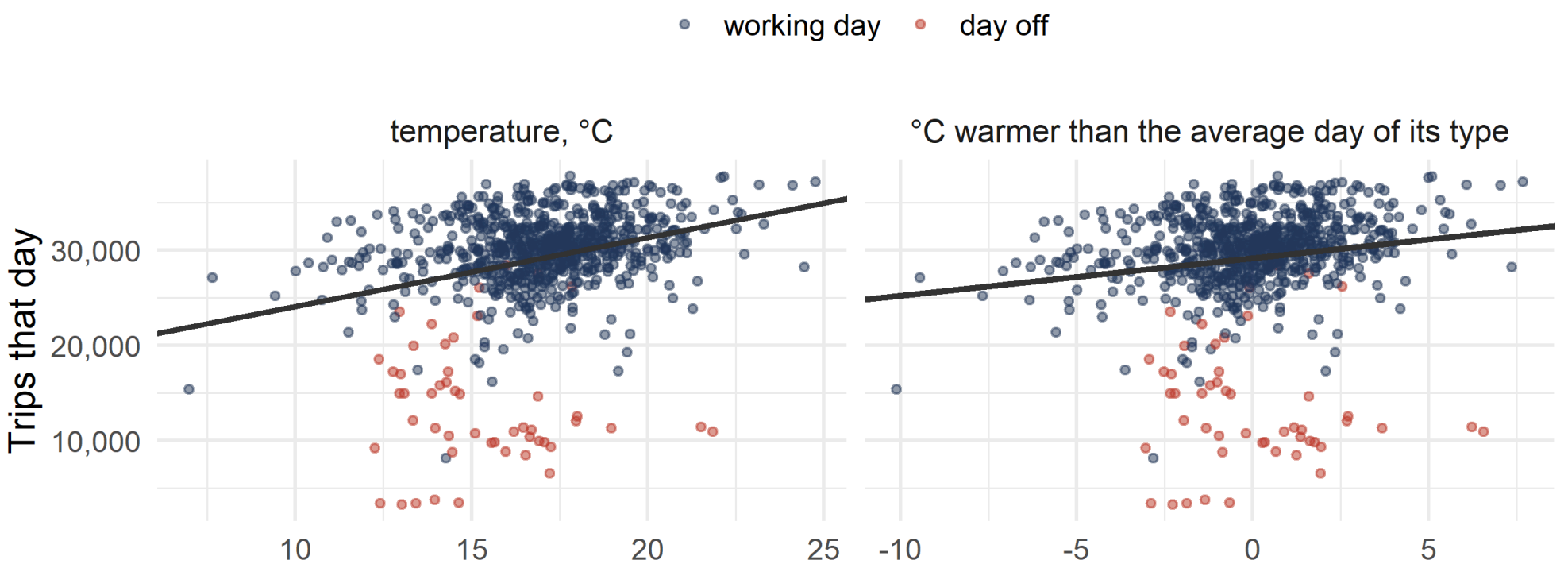


- Left: the raw PM2.5. The rainy days sit at the clean end, so the line is flatter: -61.6 trips per µg/m³.
- Right: each day against the usual PM2.5 for its rain. The rain no longer makes the clean days look empty, and the slope steepens to -87.9: the slope with rain held fixed.
- Every slope in a regression with several regressors works this way.

The same with temperature and days off

Appendix — what the days off took out

The cold-days example, with the days off held fixed in the same way.



- Left: the raw temperature. The days off sit at the cold end, so the line is steep: 724 trips per degree.
- Right: each day against the average day of its type. The days off no longer look cold, and the slope falls to 395: the slope with the days off held fixed.
- Every slope in a regression with several regressors works this way.

← Back to the rain

Appendix — is the slack hidden?

A schedule planner looks at the scheduled time only: do longer flights land earlier? What happens to the `sched_time` slope once the departure delay is held fixed?

1. Regress `flights$arr_delay` on `flights$sched_time` alone.
2. Compare its slope with the `sched_time` slope of Exercise 5, where `flights$dep_delay` is held fixed.
3. Compute the correlation between `flights$sched_time` and `flights$dep_delay`.
4. Why did the `sched_time` slope move? What does it mean for the planner?

Exercise | Start Over | Show Solution | Run Code

```
1
```

In the language of the rule: the departure delay moves the arrival almost minute for minute, and it moves a little with the scheduled time.

← Back to the rule

Appendix — collinearity: today or yesterday?

Near-copies are allowed, but they make each slope hard to pin down.

`bikes$temp_yday` is yesterday's temperature. Riders may decide on today's weather, or on what they remember from yesterday.

Today's and yesterday's temperature move together. What happens to the temperature slope, and to its standard error, when both go into the regression?

Appendix — collinearity, in R

1. Regress `bikes$trips` on `bikes$temp` and `bikes$dayoff`.
2. Add `bikes$temp_yday`.
3. Compare the slope on `temp` and its standard error in the two regressions. What happened to R^2?
4. Compute the correlation between `bikes$temp` and `bikes$temp_yday`. The first day has no yesterday, so use `use = "complete.obs"`.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — collinearity costs precision

Model	temp slope	SE of temp slope	R^2
trips on temp and dayoff	395	59	0.559
trips on temp, temp_yday and dayoff	525	90	0.562

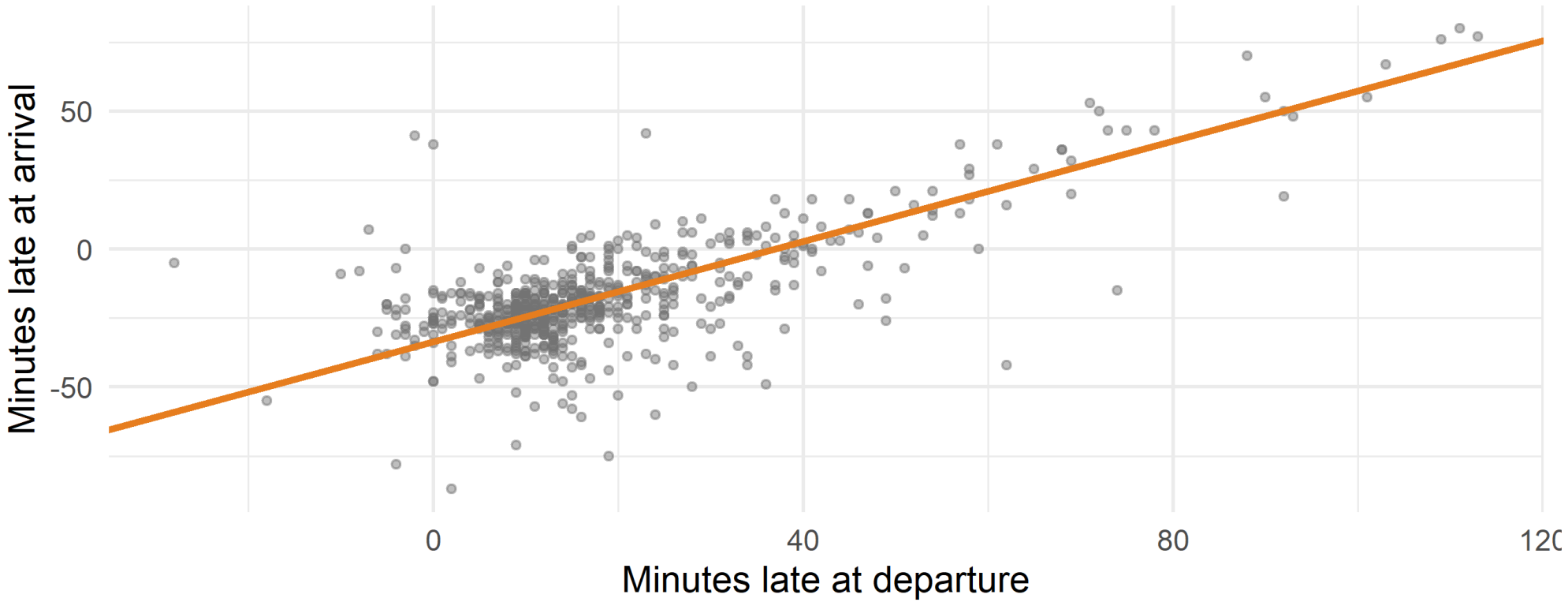
- Today's and yesterday's temperature move together (correlation 0.76). The data can hardly tell which one riders respond to: the standard error of the temperature slope grows by half, and yesterday's slope even takes the wrong sign.
- The fit barely changes: R^2 goes from 0.559 to 0.562.
- The hours column of Exercise 3 is the extreme: a perfect copy, and R drops it.

For a forecast, collinearity costs little. For saying which cause matters, it costs a lot.

← Back to full rank

Appendix — one regressor: a line

What does $\hat{\beta}$ draw? With one regressor, the fitted values lie on a line.



$\widehat{\text{arr_delay}} = -33.5 + 0.910\,\text{dep_delay}$: two numbers, one line.

600 of the 11,887 flights are drawn; the line is fitted on all of them.

← Back to the plane

Appendix — least squares by hand

Minimise $SSE(b_0, b_1, b_2) = \sum (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2})^2$ without matrices. As in W07, set each derivative to zero. First the intercept.

$$\frac{\partial SSE}{\partial b_0} = -2 \sum_i e_i = 0$$

$$e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}$$

$$n\bar{y} - n\hat{\beta}_0 - \hat{\beta}_1\, n\bar{x}_1 - \hat{\beta}_2\, n\bar{x}_2 = 0$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2$$

$$\boxed{SSE = \sum_i (\tilde{y}_i - b_1 \tilde{x}_{i1} - b_2 \tilde{x}_{i2})^2}$$

Derivative in b_0
Each squared error changes by $2e_i$ times -1.

Sum each term
$\sum y_i = n\bar{y}, \sum \hat{\beta}_0 = n\hat{\beta}_0, \sum x_{i1} = n\bar{x}_1, \sum x_{i2} = n\bar{x}_2.$

Divide by n
The fitted plane passes through the point of means.

Put $\hat{\beta}_0$ back in
With $\tilde{y}_i = y_i - \bar{y}, \tilde{x}_{i1} = x_{i1} - \bar{x}_1, \tilde{x}_{i2} = x_{i2} - \bar{x}_2$: every variable measured from its mean. Two unknowns are left.

The two slopes → ← Back to least squares

Appendix — least squares by hand: the two slopes

Write $S_{11} = \sum \tilde{x}_{i1}^2$, $S_{22} = \sum \tilde{x}_{i2}^2$, $S_{12} = \sum \tilde{x}_{i1}\tilde{x}_{i2}$, $S_{1y} = \sum \tilde{x}_{i1}\tilde{y}_i$, $S_{2y} = \sum \tilde{x}_{i2}\tilde{y}_i$.

$$\frac{\partial SSE}{\partial b_1} = -2 \sum_i \tilde{x}_{i1}\, e_i = 0$$

$$e_i = \tilde{y}_i - \hat{\beta}_1 \tilde{x}_{i1} - \hat{\beta}_2 \tilde{x}_{i2}$$

$$S_{11}\,\hat{\beta}_1 + S_{12}\,\hat{\beta}_2 = S_{1y}$$

$$S_{12}\,\hat{\beta}_1 + S_{22}\,\hat{\beta}_2 = S_{2y}$$

$$\left(S_{11}S_{22} - S_{12}{}^2\right)\hat{\beta}_1 = S_{22}\,S_{1y} - S_{12}\,S_{2y}$$

$$\boxed{\hat{\beta}_1 = \frac{S_{22}\,S_{1y} - S_{12}\,S_{2y}}{S_{11}\,S_{22} - S_{12}{}^2}}$$

Derivative in b_1
Each squared error changes by $2e_i$ times $-\tilde{x}_{i1}$.

Split the sum, name each part
$\sum \tilde{x}_{i1}\tilde{y}_i - \hat{\beta}_1 \sum \tilde{x}_{i1}^2 - \hat{\beta}_2 \sum \tilde{x}_{i1}\tilde{x}_{i2} = 0.$

Derivative in b_2
The same steps, with $\tilde{x}_{i2}$ in front.

Remove $\hat{\beta}_2$
Multiply the first equation by S_{22} and the second by S_{12}, then subtract.

Divide
$\hat{\beta}_2$ is the same with 1 and 2 swapped.

"Held fixed, inside the formula" reaches the same result from $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$. If x_1 and x_2 do not move together, $S_{12} = 0$ and $\hat{\beta}_1 = S_{1y}/S_{11}$: the one-regressor slope of W07.

← Back to least squares

Appendix — what $\mathbf{X}\boldsymbol{\beta}$ gives

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

$$\text{row } 2 \times \boldsymbol{\beta} = 1 \cdot \beta_0 + x_{21}\,\beta_1 + \cdots + x_{2k}\,\beta_k$$

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \cdots + \beta_k x_{1k} \\ \beta_0 + \beta_1 x_{21} + \cdots + \beta_k x_{2k} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \cdots + \beta_k x_{nk} \end{bmatrix}$$

$$\boxed{y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i}$$

Check the sizes
$n \times (k+1)$ times $(k+1) \times 1$ gives $n \times 1$: one number per flight.

One row times $\boldsymbol{\beta}$
Multiply each entry of the row by the matching β, then add.

Every row
The same for each flight: its own x's, the same β's.

Add $\mathbf{u}$, read row i
Row i of $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ is the regression for flight i.

The flights have $k = 2$. The Monterrey flight, 40 minutes late and scheduled for 100 minutes, has the row $(1, 40, 100)$. Its forecast is $\beta_0 + 40\,\beta_1 + 100\,\beta_2$.

← Back to the matrix form

Appendix — what is left in the errors?

OLS has picked the coefficients of the regression of `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`. Can either regressor still explain part of the errors that are left?

1. Fit the regression and save its residuals as `e`.
2. Compute the mean of `e`.
3. Compute the correlation of `e` with `flights$dep_delay`, and with `flights$sched_time`.
4. What do the three numbers say?

Exercise | Start Over | Show Hint | Run Code

```
1
```

← Back to the condition

Appendix — four flights, by hand

Four flights: minutes late at departure, scheduled time in hours, and minutes late at arrival.

Flight	Departure delay x_1 (minutes)	Scheduled time x_2 (hours)	Arrival delay y (minutes)
1	0	1	−22
2	10	1	−16
3	10	2	−20
4	20	2	−6

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 10 & 1 \\ 1 & 10 & 2 \\ 1 & 20 & 2 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} -22 \\ -16 \\ -20 \\ -6 \end{bmatrix}$$

The blue column of 1s carries the intercept.

← Back to the formula

Appendix — four flights: the steps

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 4 & 40 & 6 \\ 40 & 600 & 70 \\ 6 & 70 & 10 \end{bmatrix}$$

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \begin{bmatrix} 2.75 & 0.05 & -2 \\ 0.05 & 0.01 & -0.1 \\ -2 & -0.1 & 2 \end{bmatrix}$$

$$\mathbf{X}^\top\mathbf{y} = \begin{bmatrix} -64 \\ -480 \\ -90 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \boxed{\begin{bmatrix} -20 \\ 1 \\ -4 \end{bmatrix}}$$

How the regressors move together
Row by column: $n = 4$, $\sum x_1 = 40$, $\sum x_1^2 = 600$, $\sum x_2 = 6$, $\sum x_1 x_2 = 70$, $\sum x_2^2 = 10$.

Untangle them
The inverse: multiplied by $\mathbf{X}^\top\mathbf{X}$ it gives the identity matrix.

How each regressor moves with y
$\sum y = -64$, $\sum x_1 y = -480$, $\sum x_2 y = -90$.

Multiply
One minute late at arrival per minute late at departure, for the same scheduled time; 4 minutes earlier per extra scheduled hour, for the same departure delay. The residuals are ± 2.

← Back to the formula